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POLRAG: A RAG-LLM Framework for Policy Question Answering

Hongquan Lin[a,b], Pingping Deng[a,b], Qianshan Zhong[a,b], and Xiaoying Zhu[a,b,*]

[a]*Guangxi Key Laboratory of Machine Vision and Intelligent Control, Wuzhou University, Wuzhou, 543002 China*
[b]*Guangxi Colleges and Universities Key Laboratory of Intelligent Software, Wuzhou University, Wuzhou, 543002 China*
[*]mailto: 980189526@qq.com

***Abstract*—In the modern social governance system, policy documents are the core carriers of public services and social norms, carrying a large amount of key information. However, as the policy system continues to expand and update, users are faced with problems such as information overload and low retrieval efficiency. Traditional policy consultation models generally have pain points such as delayed response and high service costs. In response to the above challenges, this paper proposes a RAG-LLM framework for policy question answering (POLRAG) . In the knowledge base construction stage, by structurally decomposing and slicing the policy text, a domain-adapted embedding model is used to convert knowledge units into high-dimensional vectors, which are stored in a distributed vector database to achieve millisecond-level semantic retrieval. In the retrieval stage, in order to solve the ambiguity and randomness of user questions, we use LLM to identify the intention and rewrite the structure of user questions. Secondly, in order to improve the comprehensiveness and accuracy of retrieval, we adopt a multi-channel recall strategy, leveraging the BM25 algorithm and semantic similarity algorithms to filter relevant information from the knowledge base, followed by re-ranking and merging the retrieval results. In the generation phase, POLRAG innovatively incorporates historical conversation context and retrieval results into the prompt template, constructs a multi-turn interactive reasoning framework, and ultimately enables the LLM to generate responses that are accurate, compliant, and practical. In tests covering policy scenarios such as social security processing and employment subsidies, POLRAG significantly outperformed general large models in core indicators such as answer accuracy and policy matching.**

***Index Terms*—RAG; LLM; Policy question answering**

I. Introduction

With the rapid development of artificial intelligence, large language models (LLMs) have become crucial in intelligent question answering. However, issues like knowledge hallucination and insufficient domain knowledge limit their application in policy question answering [1], where understanding professional content accurately is the key. Retrieval-augmented generation (RAG) integrates LLMs with information retrieval systems [2]. By retrieving relevant information from documents and feeding it along with user queries into LLMs, RAG enables access to up-to-date knowledge, mitigates "knowledge hallucination," and improves answer accuracy.

This study proposed POLRAG, an intelligent policy question-answering framework, by integrating LLMs with text embedding and vector databases. Its main contributions are:

- Framework innovation: POLRAG combines RAG and LLM technologies, offering a new solution for government intelligent question answering by addressing knowledge update lags and accuracy issues.
- Intent analysis: An LLM-based intent recognition module converts unstructured user questions into structured requirements, ensuring clear input for retrieval and answer generation.
- Retrieval optimization: POLRAG uses a multi-way recall strategy, combining the BM25 [3] and semantic similarity algorithms to retrieve policy information. A re-ranking mechanism further refines search results, facilitating high-quality answer generation.

II. Related Work

A. Large Language Model

Since the Google research team published Attention is all you need [4], the Transformer architecture has promoted a new round of deep learning development. In particular, the advent of ChatGPT has directly triggered a new round of artificial intelligence craze and accelerated the development of generative artificial intelligence technology and AI computing power. Large language models originated from early pre-trained language models, such as BERT proposed by Devlin et al. [5] and GPT developed by OpenAI [6]–[8], which respectively adopted the encoder and decoder modules of the Transformer framework. As the field of large language models continues to flourish, new models continue to emerge. On January 20, 2025, DeepSeek company released the DeepSeek-R1 AI large model, which uses reinforcement learning to stimulate the reasoning ability of large language models and introduces multi-stage training and cold start data to further improve reasoning performance [9]. Google officially launched Gemma 3 on March 12, 2025. Based on Gemma, it adds visual understanding capabilities, expands language coverage, and optimizes long-context processing capabilities. It solves memory problems through architectural adjustments and improves the overall performance of the model [10]. On April 29, 2025, the Alibaba Tongyi Qianwen team released Qwen3, which integrates thinking mode and non-thinking mode into a unified framework. It also introduces a thinking budget mechanism that can adaptively allocate computing resources during the

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reasoning process according to the complexity of the task, balancing latency and performance [11].

B. Retrieval-Augmented Generation Technology

Large language models have shown strong capabilities in the field of natural language processing, but they still face problems such as hallucinations and lack of domain-specific knowledge. Retrieval-augmented generation technology uses large-scale external knowledge bases to enhance the semantic understanding and generation capabilities of the model, effectively alleviating some of the problems faced by large language models. Retrieval enhancement technology [12] was first proposed by Meta AI in 2020. It combines a retriever with a pre-trained sequence-to-sequence model, so that the implicit knowledge of parameter memory within the model and the knowledge base knowledge of non-parametric memory outside the model are combined to help the model retrieve generation. RAG is a model architecture that combines information retrieval technology with a language generation model. It generates answers or content by referencing information from an external knowledge base. Zheng Wang et al. [13] constructed M-RAG and introduced a multi-partition paradigm, taking each database partition as the basic unit for RAG execution, and using a novel framework that combines a large language model with multi-agent reinforcement learning to explicitly optimize different language generation tasks . At the same time, Cheng Niu et al. [14] proposed the RAGTruth corpus for in-depth analysis of word-level hallucinations, providing a solid data basis for the subsequent formulation of effective prevention strategies. Weihang Su et al. [15] constructed the DRAGIN framework, which can accurately decide when to search and what to search based on the real-time information needs of the large language model during the text generation process, making the retrieval process more intelligent and efficient. Akari Asai et al. [16] constructed the Self-RAG framework to improve the quality and authenticity of the model through retrieval and self-reflection. These continuous research and innovations have strongly promoted the development of RAG technology.

III. Proposed Model

A. Model Overview

We propose an innovative framework, RAG-LLM Framework for Policy Question Answering, called POLRAG. It is divided into three stages: knowledge base construction stage, retrieval stage, and generation stage. In the knowledge base construction phase, we mainly divide text paragraphs according to their semantics to build a structured knowledge system; in the retrieval phase, we innovatively use query optimization and multi-way recall strategies to resolve vague and arbitrary questions asked by users and improve the comprehensiveness and accuracy of retrieval. In the generation phase, we build a multi-round interactive reasoning template and inject historical dialogue information into the prompt word template. The flowchart of the POLRAG framework is shown in the figure1.

B. Build a Local Knowledge Base

In the policy question-answering system, high-quality domain-specific corpora are the key to expanding information sources and strengthening the ability of embedded models. To this end, we collected more than a thousand policy documents covering core areas such as social security and law from official government websites. In the data preprocessing stage, we use a multi-level screening mechanism to ensure data quality: first, we use regular expressions to identify and remove invalid files containing garbled characters; second, we use HTML parsing technology to automatically filter non-semantic information such as hyperlinks, charts, tables, and various embedded tags.

In the content segmentation phase, we combined the characteristics of policy documents, used “chapter and regulation completeness” and “single file block 200-300 words” as dual standards, scientifically segmented the documents, and extracted the corresponding chapter titles as file block metadata for subsequent retrieval and positioning. In the vector conversion process, the BGE (Byte Pair Encoding) embedding model was used to convert the text content into a high-dimensional numerical vector, and finally the vector data and metadata were synchronously stored in the Chroma vector database, successfully building an efficient and accurate local vector knowledge base.

C. Multi-channel Recall Retrieval Strategy

In the information retrieval system, in order to cope with the diversity and ambiguity of user queries and improve the comprehensiveness and accuracy of retrieval, we innovatively adopt a multi-way recall retrieval strategy, integrating the two core technical paths of semantic similarity retrieval and keyword retrieval to achieve complementary advantages.

Semantic similarity retrieval leverages the vector space model and Sentence Transformer framework to map text into 768-dimensional semantic vectors. Once a user enters a query, the system converts it into a semantic vector via a pre-trained encoder and applies L2 normalization to standardize vector scale, enhancing cosine similarity calculation. For data management, the Chroma vector database stores policy document vectors, with an HNSW graph index built to cluster vectors hierarchically. This structure accelerates nearest neighbor searches and reduces retrieval latency. During querying, the system calculates cosine similarity between the query vector and database vectors, selects the top k closest documents as candidates, and converts vector distances to similarity scores using the 1-distance formula for semantic relevance evaluation.

Keyword search employs the BM25 algorithm, a classic probabilistic model leveraging the bag-of-words approach. When a user submits a query, the system first uses the jieba tokenizer to split the text into tokens. Then, the BM25 model calculates relevance scores between the token list and each policy document based on a pre-built corpus index. During index construction, the system parses policy documents into word collections and creates an inverted index. For retrieval, it ranks documents by scores, selects the top k, and combines

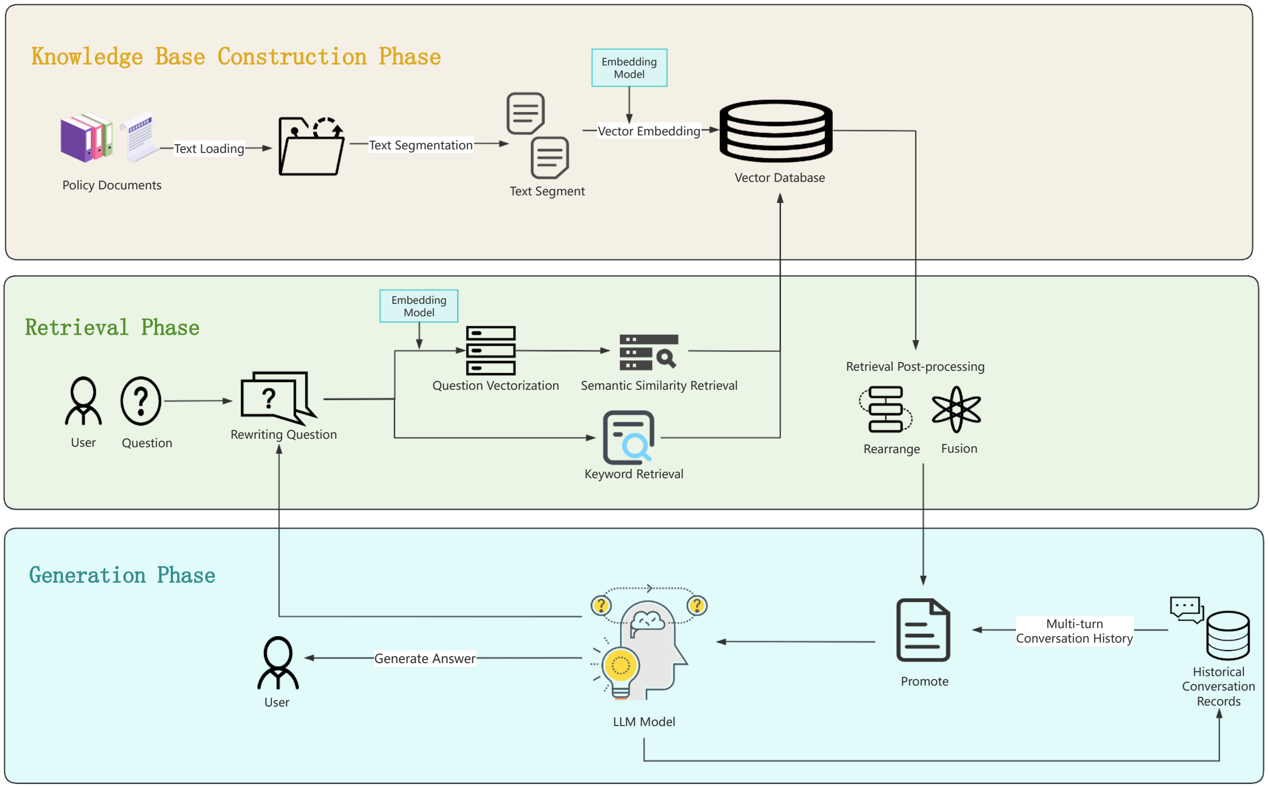

Fig. 1. Flowchart of the POLRAG Framework.

document metadata (e.g., release time, source) with standardized BM25 scores. Only documents with scores exceeding 0 are retained, effectively filtering out noise.

The multi-path recall strategy deeply integrates the two retrieval methods. Semantic retrieval excels at capturing implicit semantic connections, addressing recall issues in fuzzy queries, while keyword retrieval precisely matches user-entered explicit words, ensuring accurate retrieval of professional terms. By using a dynamic weight allocation and result fusion mechanism, this strategy boosts the retrieval system's recall rate and accuracy. It enables more efficient and comprehensive responses to complex user retrieval needs.

D. Query Optimization

In order to significantly improve the accuracy of retrieval, this system innovatively built a large model-driven structured question rewriting mechanism. This strategy converts users' natural language questions into standardized and normalized structured expressions, laying a solid foundation for the subsequent retrieval process.

The underlying system relies on the large language model deployed by the Ollama service to deeply tap into its excellent semantic parsing and text generation capabilities. When a user submits a question, the system immediately starts the customized prompt engineering module and sends precise instructions to the large language model: on the one hand, the task goal is to convert the original question into a structured expression suitable for policy retrieval; On the other hand, the output format is strictly regulated, requiring it to be presented in the form of "structured expression: subject: xxx; elements: xxx, xxx, xxx", and presetting typical search element categories such as application conditions, reimbursement process, and required materials. At the same time, through the guidance of multiple examples that fit the actual scenario, the model is helped to accurately grasp the task requirements, and finally output a structured expression that is highly adapted to the search scenario, which improves the search efficiency and accuracy in all aspects.

E. Prompt Engineering

In order to ensure that the large language model (LLM) can output professional and accurate answers in government consultation scenarios, the system innovatively builds a structured prompt word engineering system. Through scientifically designed input frameworks and guidance strategies, it ensures that the answer content deeply integrates key elements such as policy basis citations and step-by-step guidance. The system adopts a three-stage progressive architecture of "system role setting + policy context construction + answer guidance and example supplementation". First, the system assigns LLM the identity of "Government Policy Consulting Assistant" through system messages, and strictly regulates the answer standards, framing the direction of professional answers from the source; then, based on the search results, the system organizes the recalled document information in the policy database in a structured manner, using the unified format of "File Name, Clause Number: Specific Content" to provide LLM with authoritative and accurate knowledge support, ensuring that the answers are based on the law; Finally, the system intelligently analyzes the search content to generate answer guidelines, indicating the key points and logical framework of the answer, and combines the Few-Shot learning strategy to introduce typical question and answer templates such as high-level talent certification material consultation to assist LLM in clearly grasping the expected output format. Finally, the system organically integrates role settings, user questions, policy fragments, answer guidelines and examples to form a complete user message, driving LLM to output professional and standardized government consultation answers.

IV. EXPERIMENT

This experiment is based on an Intel 64-core CPU and 16G memory computing environment, equipped with an NVIDIA GTX 1660Ti graphics card (6G memory), using CUDA 12.8 accelerated computing, and installing the pytorch deep learning framework to provide stable hardware and software support for model training and testing.

A. Evaluation Metrics

To accurately evaluate model performance, this study constructs a three-dimensional system covering accuracy, completeness, and normativeness, as shown in the tableI. The accuracy dimension checks the match between model output and authoritative policy texts, verifying policy clause citations and business content authenticity. Completeness evaluation focuses on whether the model addresses all key elements in user questions, like business procedures and application requirements. For normativeness, the model must adhere to government professional terminology standards and use structured formats (e.g., bullet points, tables) to ensure clarity. By combining weighted scores from these three dimensions, the system objectively assesses the model's effectiveness in government policy question-answering.

B. Ablation Study

In order to deeply explore the specific contribution of each core module to the system performance, this study carefully designed a layered ablation experiment, and built a comparison model with different configurations by gradually removing key modules. The large model used in this study is Qwen3:1.7B. The specific operations are as follows:

Step 1: Remove all recall and query optimization modules: Get a general large language model that only retains basic capabilities, named BaseLLM, as a benchmark for performance comparison.

Step 2: Keep the semantic similarity retrieval module: On the basis of BaseLLM, only retain the Retrieval-Augmented Generation mechanism based on semantic similarity retrieval, build the SemanticRAG model, and verify the performance improvement effect of semantic retrieval.

Step 3: Add query optimization module: On the basis of the SemanticRAG model, retain the semantic similarity retrieval function, add the query optimization module, and build the OptiRAG model, aiming to analyze the impact of query optimization on retrieval accuracy.

Step 4: Superimpose keyword retrieval module: On the basis of the SemanticRAG model, add keyword retrieval capabilities to form the HybridRAG model, and evaluate the effectiveness of the semantic and keyword hybrid retrieval strategy.

Step 5: Full function model: Finally, retain the query optimization and multi-way recall modules to generate the FullRAG model, as an experimental group with complete functions, and compare the performance with the above models.

Through a hierarchical ablation experiment, the system compares the performance of five models across the three dimensions of accuracy, completeness, and standardization. This approach enables a quantitative evaluation of the contributions made by the multi-way recall and query optimization module to the overall system performance. To conduct this evaluation, the five models were tasked with responding to 50 questions, after which three experts were invited to score the responses. The scoring results are presented in Table II.

Scoring results reveal that different module configurations significantly impact system performance. BaseLLM, lacking retrieval and optimization modules, scores lowest across all dimensions (average 6.1), highlighting the value of semantic retrieval in government policy question answering. SemanticRAG, incorporating semantic similarity retrieval, boosts the average score to 7.4, confirming its effectiveness in improving policy content matching. OptiRAG, adding a query optimization module to SemanticRAG, raises the average score to 8.1, further enhancing performance in all dimensions. HybridRAG combines keyword and semantic retrieval, achieving an average score of 8.2, showing that hybrid retrieval broadens information coverage, though its impact on standardization requires further verification. FullRAG, integrating all modules, reaches an average score of 9 and tops in every dimension, demonstrating that multi-module collaboration maximizes model performance. Overall, each module contributes positively, and their combined use optimizes accuracy, completeness, and standardization in policy question answering, guiding system improvement.

C. Case Study

As shown in Table III, BaseLLM only gives broad descriptions of policies like employment internship and job-seeking subsidies. Its content is scattered and vague, with statements such as "consult the local human resources and social security department", merely skimming the surface of policy existence without detailing subsidy conditions or calculations. This offers limited reference to local users and

TABLE I
EVALUATION METRICS

Evaluation Metrics	Scoring Standards	Score range
Accuracy	10 points: Complete citation of policy basis, and the answer content is fully consistent with the current policy. 8–9 points: There are minor omissions, but they do not affect the core content. 6–7 points: Key information is missing or there are errors in citation. 0–5 points: The answer is wrong or there is no policy basis.	0–10 points
Completeness	10 points: Fully cover the key elements of the problem, and the descriptions of multiple situations are complete. 8–9 points: Missing secondary elements. 6–7 points: Key elements are missing. 0–5 points: The core problem is not answered.	0–10 points
Normativeness	10 points: Government affairs terms are accurate, and the format is clear and standardized. 8–9 points: There are a small number of expression or format flaws. 6–7 points: Ambiguous expressions or chaotic format. 0–5 points: Expression errors or no format.	0–10 points

TABLE II
COMPARISON OF MODEL EVALUATION RESULTS

Model Name	Accuracy			Completeness			Normativeness			Average score
	Expert 1	Expert 2	Expert 3	Expert 1	Expert 2	Expert 3	Expert 1	Expert 2	Expert 3	
BaseLLM	5	5	5	6	6	7	7	7	6	6.1
SemanticRAG	6	8	8	6	7	8	8	8	6.7	7.4
OptiRAG	8	8	9	6	9	8	8	8	7.4	8.1
HybridRAG	8	8	8	7	9.5	8	8	8	7.7	8.2
FullRAG	9	9	9	8	9.5	9	9	9.5	8.7	9

TABLE III
MODEL RESPONSE EXAMPLE1

Model Name	The question is: I am a fresh graduate this year. What are the employment policies for college graduates like us?
BaseLLM	According to the policy basis retrieved by the system, the employment support policies for college graduates mainly include the following: 1. Employment Probation Subsidy: Eligible college graduates can apply for employment probation subsidies. For specific standards and procedures, please consult the local human resources and social security department. 2. Job-Hunting and Entrepreneurship Subsidy: College graduates who meet the national regulations can apply for the job-hunting and entrepreneurship subsidy. The subsidy amount and application conditions are specified in the *Notice on Further Supporting Employment and Entrepreneurship of College Graduates* (Ren She Bu Fa [2021] No. 17). 3. Flexible Employment Support: College graduates are encouraged to engage in flexible employment. Relevant support policies include vocational skills training subsidies, social security subsidies for flexible employment, etc. For specific rules, please contact the local employment service center. 4. Residence Registration and File Management: Eligible college graduates can apply for residence registration, and file management services are coordinated by the human resources and social security departments. Please supplement specific regions or policy directions for further detailed information. For specific operation guidelines, it is recommended to consult the local human resources and social security bureau in person.

fails to guide practical operations, requiring users to consult multiple times and increasing business costs.

In contrast, as shown in Table IV, based on Guangxi's policies, FullRAG refines the business chain, clarifying details like "job-seeking subsidies for low-income family graduates are 80% of the minimum wage". It marks policy terms, anchors local documents, and adapts to regional needs. By providing online portals, review time limits, and predicting information requirements, it turns policies into actionable services, solving users' practical problems. This showcases the value of multi-module collaboration in government Q&A, better meeting the demands for precision, efficiency, and convenience. The two models significantly differ in content, information depth, policy adaptability, and user service value.

V. CONCLUSION

This study aims to solve the core problems of lagging knowledge update and insufficient answer accuracy in government policy question-answering scenarios by constructing an intelligent policy question-answering framework POLRAG based on the RAG architecture, which optimizes the entire process from user intent analysis to accurate answer generation. The LLM-based intent recognition and structured rewriting module is designed to convert users' fuzzy and random natural language questions into standardized retrieval requirements through semantic analysis. The multi-way recall strategy of the BM25 algorithm and the semantic similarity algorithm are used to achieve comprehensive and accurate retrieval of policy information. Experimental verification shows that compared with the basic model, POLRAG significantly improves the accuracy, completeness, and standardization of policy questions and answers. This framework provides a feasible technical path for the government intelligent question-answering system, helping government services to evolve towards more accuracy, efficiency, and convenience.

TABLE IV
MODEL RESPONSE EXAMPLE2

Model Name	The question is: I am a fresh graduate this year. What are the employment policies for college graduates like us?
FullRAG	Instructions on Employment Policies for College Graduates In accordance with the "Administrative Measures for Employment Subsidy Funds of Guangxi Zhuang Autonomous Region" and related policy documents, matters such as employment subsidies, probation management, and fund application for college graduates shall be explained as follows: I. Eligible Objects College graduates (including those in the graduation year) College graduates from low-income families, zero-employment families, and families under anti-poverty monitoring. Disabled college graduates and those who have obtained national student loans (in accordance with Article 10 of the "Expenditure Scope and Standards for Employment Subsidy Funds of Guangxi Zhuang Autonomous Region"). Individual industrial and commercial households that hire college graduates (in accordance with Article 10 of the "Expenditure Scope and Standards for Employment Subsidy Funds of Guangxi Zhuang Autonomous Region"). II. Application Methods and Handling Channels 1. Application Methods Online application through the autonomous region's "Digital Smart Human Resources and Social Security" information system (or data sharing with human resources and social security, and finance departments). Individual industrial and commercial households need to be reviewed by the local human resources and social security department or finance department. 2. Handling Channels Individuals: Log in to the official website of the "Guangxi Zhuang Autonomous Region Department of Human Resources and Social Security" or the "Digital Smart Human Resources and Social Security" platform. Enterprises/individual industrial and commercial households: Contact the local human resources and social security department window or the finance fund distribution unit. III. Subsidy Amounts and Standards 1. Job-Hunting Subsidies College graduates from low-income families, zero-employment families, and families under anti-poverty monitoring: 80% of the monthly minimum wage standard in Class I areas of the region in the previous year (e.g., 1,500 yuan/person). Disabled college graduates and those with national student loans: The same standard. 2. Employment Expansion Subsidies Enterprises and individual industrial and commercial households that hire college graduates and sign labor contracts of more than 1 year and pay social insurance: 1,500 yuan per person. 3. Employment Subsidies Enterprises that organize rural laborers and registered unemployed persons for employment and sign labor contracts of more than 3 months: no more than 1,000 yuan per person. IV. Application Scope and Handling Time Limits 1. Application Scope Employment of key groups (college graduates, low-income families, anti-poverty monitoring objects, etc.). Subsidy policies for enterprises and individual industrial and commercial households that hire employees. 2. Handling Time Limits Review of application materials and fund allocation: shall be completed within the graduation year, and the specific time shall be implemented in accordance with local policies. Subsidies for individual industrial and commercial households shall be issued after filing with the human resources and social security department at the enterprise registration place. V. Policy Basis 1. "Administrative Measures for Employment Subsidy Funds of Guangxi Zhuang Autonomous Region" (Gui Ren She Fa [2019] No. 16). 2. "Administrative Measures for Employment Probation of Guangxi Zhuang Autonomous Region" (Gui Ren She Fa [2019] No. 16). 3. "Expenditure Scope and Standards for Employment Subsidy Funds of Guangxi Zhuang Autonomous Region" (Gui Cai She [2020] No. 12).

REFERENCES

[1] Q. Xiaolin, G. Xu, L. Dicheng, and X. Haiwen, "Survey and prospect of large language models," *Journal of Computer Applications*, vol. 45, no. 03, pp. 685–696, 2025.

[2] L. Xueying, Y. Jing, L. Bo, S. Xiaoguo, and Z. Yuying, "A survey of retrieval-augmented generation based large language model," *Computer Engineering and Applications*, pp. 1–31.

[3] S. Robertson, H. Zaragoza *et al.*, "The probabilistic relevance framework: Bm25 and beyond," *Foundations and Trends® in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[5] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, volume 1 (long and short papers)*, 2019, pp. 4171–4186.

[6] A. Radford, K. Narasimhan, T. Salimans, I. Sutskever *et al.*, "Improving language understanding by generative pre-training," 2018.

[7] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever *et al.*, "Language models are unsupervised multitask learners," *OpenAI Blog*, vol. 1, no. 8, p. 9, 2019.

[8] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.

[9] D. Guo, D. Yang, H. Zhang, J. Song, R. Zhang, R. Xu, Q. Zhu, S. Ma, P. Wang, X. Bi *et al.*, "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning," *arXiv preprint arXiv:2501.12948*, 2025.

[10] G. Team, A. Kamath, J. Ferret, S. Pathak, N. Vieillard, R. Merhej, S. Perrin, T. Matejovicova, A. Ramé, M. Rivière *et al.*, "Gemma 3 technical report," *arXiv preprint arXiv:2503.19786*, 2025.

[11] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv *et al.*, "Qwen3 technical report," *arXiv preprint arXiv:2505.09388*, 2025.

[12] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.

[13] Z. Wang, S. Teo, J. Ouyang, Y. Xu, and W. Shi, "M-rag: Reinforcing large language model performance through retrieval-augmented generation with multiple partitions," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 1966–1978.

[14] C. Niu, Y. Wu, J. Zhu, S. Xu, K. Shum, R. Zhong, J. Song, and T. Zhang, "Ragtruth: A hallucination corpus for developing trustworthy retrieval-augmented language models," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 10 862–10 878.

[15] W. Su, Y. Tang, Q. Ai, Z. Wu, and Y. Liu, "Dragin: Dynamic retrieval augmented generation based on the real-time information needs of large language models," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 12 991–13 013.

[16] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-rag: Learning to retrieve, generate, and critique through self-reflection," in *International Conference on Learning Representations*, 2024.